

 END-TO-END ANALYTICS SOLUTION

ShopSphere Retail Data Analytics

Retail Sales Analytics Using Excel, SQL, Power BI, and Python

CONTEXT & CHALLENGE

Problem Statement



Retail Market Reality

Competitive markets require continuous sales tracking and customer behavior understanding.



Growth Identification

Spot hidden opportunities and operational bottlenecks from raw transactional data.



Multi-Tool Approach

Build an end-to-end pipeline driving seamless insights generation from raw logs.

ARCHITECTURE

Main Objective & Pipeline



Excel Cleaning

Data preprocessing & validation



SQL Storage

Querying & relational modelling



Python EDA

Advanced charts & stats



Machine Learning

Customer segmentation (RFM)



Power BI Dashboards

Interactive business reporting



ETL Integration

End-to-end data pipeline flow

PHASE 01

Excel: Data Cleaning & Preparation

Standardization & Duplicates

Standardize headers to snake_case and remove redundant duplicate rows across customer tables.

Missing Values & Placeholders

Fill missing ZIP codes and phone numbers with standardized "NA" or placeholder strings.

Calculated Fields

Compute total_price as $(\text{list_price} \times \text{quantity}) - \text{discount}$ in order items.

Lookups & Pivot Tables

Merge cross-table data via VLOOKUP/XLOOKUP and build sales category pivot summaries.

PHASE 02

SQL: Database Management & Querying



ERD Table Creation

Replicate exact ER diagram structures with rigorous primary and foreign key constraints.



Aggregations & Joins

Utilize JOIN, GROUP BY, and aggregate functions to synthesize complex metrics.



Top Seller Queries

Extract top 5 selling products by quantity using ORDER BY and LIMIT clauses.

PHASE 03

Python & Pandas: EDA & RFM Analysis

SQL Data Loading

Pull relational tables into Pandas DataFrames using `pandas.read_sql()`.

Exploratory Data Analysis

Execute `df.describe()`, `info()`, and `value_counts()` coupled with Seaborn plots.

RFM Metric Computation

Calculate Recency, Frequency, and Monetary values to score customer behavior.

Segmentation Export

Export ML-driven `customer_segments` tables back to SQL for reporting integration.

PHASE 04

Power BI: Visualization & Dashboarding



Data Transformation

Connect relational tables and formulate custom DAX measures for deep KPI tracking.



Interactive Slicers

Deploy dynamic charts, cards, tables, and slicers to dissect performance variables.



Actionable Insights

Analyze sales trends, store performance, staff revenue, and customer behaviors.

CONCLUSION

Project Results & Deliverables



Fully cleaned, relational dataset stored in SQL



Insightful exploratory analytics using Python



Segmentation-based insights using Machine Learning



Interactive executive dashboards in Power BI